AI-Driven Career Mentorship Using Multi-Agent Systems and LLMs: A Modular Implementation with Gemini and LangGraph

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***Abstract*—Many individuals struggle to find personalized career guidance that aligns with their current skills, experience, and evolving job market demands. Traditional mentorship is often time-consuming, inconsistent, or inaccessible. Existing platforms provide generic suggestions rather than dynamic, skill-based career planning aiming towards specific goals. Resume feedback is rarely tailored to real job roles. And applying to jobs remains manual and repetitive. This system provides an intelligent, interactive mentorship experience that helps users understand where they currently stand, what roles they can grow into, and how to get there. It analyzes a user's resume, maps their skills to a curated career path, generates a customized career plan, explores current job market trends, and creates a job application including a tailored cover letter automatically. Additionally, the system includes an integrated multimodal interview assessment module that evaluates user responses through text, voice, and facial emotion analysis, providing personalized feedback based on sentiment scoring and answer relevance. The user interacts through a friendly interface and Chat-Bot that offers guidance and mentorship throughout the journey.**

***Keywords* - AI Career Mentorship, Multi-Agent Systems, LangGraph Orchestration, LLM-Based Resume Analysis, Multimodal**

I. Introduction

Choosing a career path or transitioning into a new role is often a complex and overwhelming process. Many individuals, especially students, recent graduates, or early-career professionals lack access to structured mentorship or personalized guidance. Traditional career counseling is either expensive, generic, or simply unavailable to large segments of the population. As a result, people often make career decisions without fully understanding the skill gaps they need to address, the roles they're best suited for, or the evolving demands of the job market. In recent years, artificial intelligence (AI) and large language models (LLMs) like those developed by OpenAI and Google have shown great promise in simulating human conversation and processing unstructured data such as resumes, job descriptions, or user input. These systems are capable of understanding language, reasoning across documents, and generating context-aware responses, making them well-suited for tasks that traditionally required human mentors or advisors. This research project presents a system designed to fill the gap in accessible and personalized career mentorship. The primary objective is to offer a guided, AI-driven experience where users can receive individualized career advice by simply uploading their resume. The system analyzes the user's background, identifies the most suitable role within a predefined career track, and recommends a step-by-step career plan to reach their next professional milestone. It also conducts a real-time analysis of job trends in the target industry and location, and goes a step further to assist in job applications by tailoring resumes and generating personalized cover letters. By bringing all of these components together — resume parsing and analysis, skill mapping, career planning, job trend insights, and application generation — encompassed into one unified system of mentorship support, this project aims to democratize high-quality career mentorship using AI, making it available and effective for users regardless of their location, budget, or background.

II. Related Work

Recent advancements in artificial intelligence (AI) and large language models (LLMs) have catalyzed the development of intelligent tutoring and mentorship systems across educational and professional domains. A number of research efforts have explored the potential of AI-driven platforms to simulate mentorship, provide career guidance, and support skill development. This section reviews key contributions in this space, emphasizing their methodologies, limitations, and how the present study builds upon them.

Bagai and Mane [1] introduced a conceptual framework for an AI-powered mentorship platform designed to enhance professional development through real-time guidance, skill evaluation, and personalized support. Their work, MentorAI, offers a high-level design using natural language comprehension and user profiling but remains largely theoretical. While it outlines the importance of integrating AI for mentorship, it lacks a concrete implementation pipeline and omits real-time feedback mechanisms or dynamic job market integration—gaps

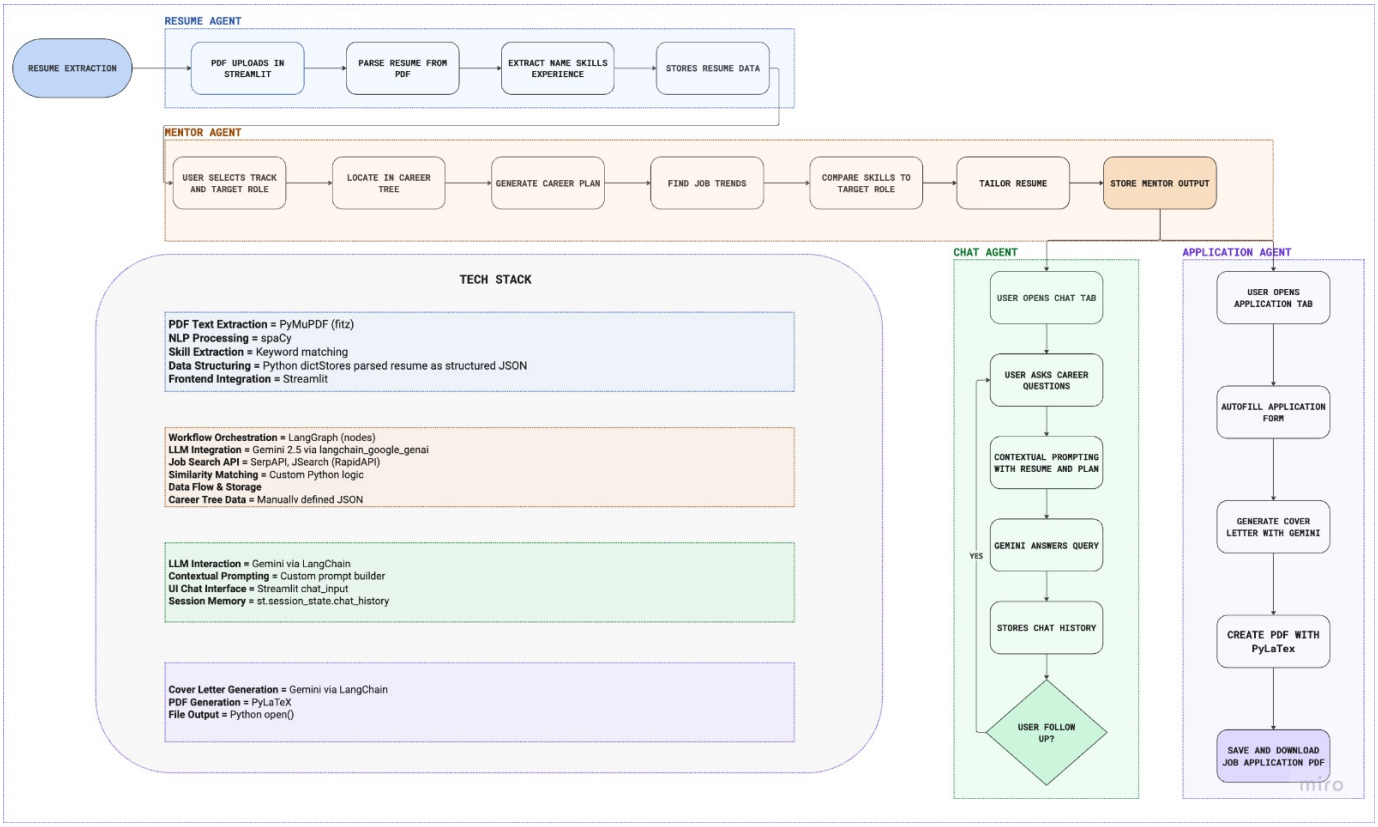

Fig. 1. Pipeline of the AI-powered mentorship system

which our modular, LangGraph-based system addresses with functional agents and resume-driven planning.

The Steve system developed by Renji et al. [2] presents a more practical application, offering a chatbot-based platform that uses LLMs to analyze career trajectories and generate resume recommendations. Although Steve supports user interaction and resume tailoring, its reliance on chunked parsing and template-based prompting limits adaptability. In contrast, our system employs deterministic resume parsing using spaCy and PyMuPDF, structured prompt engineering with Gemini, and a fit-score mechanism based on skill-tree mapping, resulting in higher personalization and clarity in role recommendations.

The growing interest in agentic workflows is well-documented by Kamalov et al. [3], who review how multi-agent AI systems can enhance learning environments by improving tool use, feedback, and task-specific adaptation. Their review underscores the pedagogical value of goal-oriented, modular agents, which aligns with our own use of discrete agents (Resume, Mentor, Search, and Application) connected via a shared state graph. However, their work focuses on general educational impact and lacks a domain-specific application, which our system provides by targeting the job-seeking process with measurable outputs such as resume tailoring, career planning, and job search integration. CyberMentor, introduced by Wang, Zhou, and Chen [4], is another relevant system that applies LLMs to cybersecurity education by leveraging retrieval-augmented generation (RAG) to deliver contextualized learning and mentorship. Their platform focuses on addressing student needs through tailored guidance and domain-specific feedback, but it is narrowly scoped to cybersecurity and emphasizes academic learning more than professional readiness. Our system generalizes the mentorship experience across technical career paths, such as Software Engineering and DevOps, and incorporates resume-to-role matching and real-world job application generation.

Zha et al. [5] propose Mentigo, an intelligent agent designed to assist students in creative problem-solving (CPS) tasks. Mentigo employs agentic logic to support classroom interactions, manage tasks, and deliver feedback. The system demonstrates the value of agent-based mentoring, especially in educational contexts requiring dynamic guidance. Unlike Mentigo, however, our system extends agentic support to career mentorship beyond the classroom, using a structured JSON-based career tree and real-time job data to guide users from their current role to aspirational positions.

The LLM-powered multi-agent framework proposed by Wang et al. [6] introduces GenMentor, an intelligent tutoring system that aligns personalized learning tasks with learners' goals using fine-tuned LLMs and optimization strategies. Their approach demonstrates the strength of modularity and shared learner profiles, principles which our system also incorporates through LangGraph's DAG-based orchestration.

Candidate	Target Role	Keyword Match (%)	Fit Score (0–100)	Interpretation
A	Frontend Developer	75%	78	High Compatibility
B	DevOps Engineer	75%	81	Excellent Fit
C	ML Intern	75%	83	Strong Match
D	Backend Engineer	75%	80	Well-Aligned
E	Prompt Engineer	60%	72	Partial Skill Alignment

TABLE I
CANDIDATE ROLE FIT ANALYSIS

However, while their work is anchored in educational tutoring, our system focuses on career navigation, resume evaluation, and automated application generation using PyLaTeX and Gemini—demonstrating real-world applicability.

Lastly, Dong et al. [7] explore how Knowledge Graph-Enhanced Retrieval-Augmented Generation (KG-RAG) can improve AI tutoring by capturing structured relationships between concepts. While KG-RAG enhances coherence and relevance in responses, it primarily supports domain knowledge retrieval and lacks resume analysis or job market awareness. Our system instead integrates live job data via SerpAPI and JSearch, ensuring that recommendations are not only educationally sound but also aligned with current industry demand.

In summary, while prior works have contributed substantially to the development of AI mentorship systems—whether focused on education, creative learning, or career progression—most suffer from domain constraints, lack of integration across the end-to-end mentorship pipeline, or limited real-world applicability. The system presented in this paper synthesizes the strengths of agentic frameworks, LLM reasoning, and real-time labor market intelligence to provide a holistic and deployable AI-driven career mentorship experience.

III. METHODOLOGY

The AI-driven career mentorship system is constructed as a modular, multi-agent pipeline designed to deliver end-to-end personalized career guidance. Its architecture combines deterministic logic, large language model reasoning, and real-time web-based interaction to create a seamless user experience. The core methodology centers on decomposing the mentorship process into a sequence of specialized tasks—each delegated to a dedicated intelligent agent—and orchestrating these tasks through a directed acyclic graph (DAG) defined using the LangGraph framework. This approach facilitates both modularity and scalability while allowing individual agents to be independently extended or replaced.

A. LangGraph: Agent Orchestration Framework

At the core of the system lies LangGraph, a Python framework that models each agent as a node in a declaratively defined state machine. This decision was grounded in the need for a flexible, deterministic orchestration method capable of maintaining state across multiple agents. Unlike monolithic chains of prompts or purely event-driven API pipelines, LangGraph offers a middle ground: it combines the clarity of a directed graph with the ability to mutate and propagate shared state across nodes. This architecture is particularly effective in career mentorship scenarios where early-stage outputs—such as parsed resume data or matched career roles—must be reused and expanded upon by downstream processes such as skill gap analysis or cover letter generation. Each transition in the LangGraph is deterministic, allowing for traceable logic flow and reproducibility, which is critical for system debugging and research evaluation.

B. Resume Agent: Parsing and Information Extraction

The initial node in the graph, the Resume Agent, is responsible for parsing and extracting structured information from uploaded resume files. Given that resumes are typically submitted as multi-page PDFs with varied layouts and formats, a robust document parsing tool was essential. For this reason, PyMuPDF (fitz) was selected due to its accuracy in preserving document formatting and line structure during text extraction. Unlike optical character recognition (OCR) tools, which can introduce noise, PyMuPDF works directly with PDF text streams and is capable of handling complex document structures. Once raw text is extracted, it is passed into a natural language processing pipeline powered by spaCy using the en_core_web_sm model. This model, although lightweight, provides reliable named entity recognition (NER) capabilities which are further enhanced by custom rule-based token filters to extract skills, experiences, and contact information from the resume. The combination of deterministic regex filtering and spaCy's probabilistic tagging strikes a balance between precision and speed, ensuring that the resume is transformed into a normalized dictionary object with fields such as name, skills, experience, and projects.

C. Mentor Agent: Role Matching and Skill Mapping

After resume parsing, the output is passed to the Mentor Agent, which is responsible for identifying the most appropriate role within a predefined JSON-based career tree. This career tree is a hierarchical structure representing multiple industry tracks such as Software Engineering, Machine Learning, DevOps, and Frontend Development. Each node in the tree corresponds to a job role with associated required skills and level-based progression logic. The design of the tree was influenced by common role hierarchies observed in industry job ladders, such as Intern $\rightarrow$ Junior $\rightarrow$ Mid-Level $\rightarrow$ Senior $\rightarrow$ Lead. The user's current position in the tree is determined by calculating the intersection between their parsed skills and the required skillset of each role. This overlap matching is implemented as a deterministic comparison, offering explainability and transparency—two qualities that are often lacking in LLM-based embedding similarity methods. The rationale for using direct skill overlap instead of vector embeddings

(such as cosine similarity between sentence representations) lies in the need for traceability and reproducibility. By explicitly comparing skill tokens, the system can provide concrete explanations for why a user qualifies for a certain role or what skills they are missing to advance. In contrast to embedding-based methods, this system uses token-level keyword matching between extracted resume skills and predefined role-specific skills from the career tree. The fit score is then computed as a weighted sum of keyword match percentage and role relevance heuristics (e.g., internship vs. senior level). This enhances both explainability and alignment with real job role hierarchies. Table 1 demonstrates the fit score calculation across five user profiles mapped to distinct target roles.

D. Career Planning and Resume Tailoring via Gemini

Once the current and target roles are determined, the system constructs a personalized career advancement plan using Gemini 2.5 Flash, Google's multimodal LLM accessed through LangChain's langchain_google_genai interface. This plan outlines the next three most important steps the user should take to reach the next level in their career hierarchy. The plan generation prompt is built using a structured few-shot example combined with role and skill data from the career tree and the user's resume. These few-shot examples are dynamically templated to reflect the domain (e.g., "ML Engineering" or "DevOps") to maintain relevancy. The use of Gemini Flash was chosen for its low latency and cost-effectiveness, delivering high-quality responses while remaining suitable for deployment in real-time systems. Additionally, this LLM is used to generate a tailored resume variant aligned to the selected role. By prompting Gemini with structured resume data and the career tree's skill expectations, the model rewrites the original resume in a way that emphasizes relevant experiences, skills, and projects, thereby improving alignment with the user's career goals and target job descriptions.

E. Search Agent: Real-Time Job Market Intelligence

The Search Agent serves to bridge the gap between abstract career planning and the realities of the job market by aggregating real-time job listings. This is accomplished using two APIs: SerpAPI and JSearch via RapidAPI. SerpAPI was chosen for its reliable access to Google Jobs and its flexible query structure, while JSearch complements it by offering deeper employer metadata and alternative job description formats. The listings retrieved by both APIs are parsed, unified into a shared structure, and summarized using Gemini to present the top five job openings in a natural language format. This summary is not only displayed to the user but is also injected into the memory context of the chatbot interface for subsequent user interactions. Real-time trend awareness is essential in mentorship systems, and this design ensures that users are exposed to the most relevant opportunities as they emerge in their chosen field and location.

F. Application Agent: Document Generation and Export

The Application Agent finalizes the mentorship process by generating a complete job application bundle, including a polished resume and a personalized cover letter. The cover letter is created using Gemini, which takes as input the parsed resume, selected job title, career goals, and real job descriptions from the Search Agent. The resulting output is a coherent, professional cover letter that reflects the user's journey and motivation. To preserve formatting and exportability, both the tailored resume and the cover letter are rendered into a single PDF using PyLaTeX, a Python interface to the LaTeX typesetting system. This choice was driven by the need for high-fidelity document formatting that matches industry standards used by hiring managers and applicant tracking systems (ATS). Unlike plain text or markdown exporters, PyLaTeX ensures consistent spacing, font styling, and section layout, thereby enhancing the user's chances of passing resume screening software.

G. Streamlit Frontend: User Interface and Session Management

All components of the system are accessible through a responsive frontend built with Streamlit, chosen for its simplicity and rapid deployment capabilities in machine learning applications. The Streamlit interface includes three core tabs:

1) Resume Upload and Mentorship Selection
2) Application Material Generation
3) Conversational AI Chat Window

Each tab interacts with the LangGraph backend through shared session state variables maintained in st.session_state, which track uploaded resume data, agent outputs, and ongoing chat history. This architecture enables a fluid user experience where different stages of the mentorship process are logically separated yet interconnected.

H. Multimodal Interview Agent: Integration and Analysis

A new Multimodal Interview Agent was introduced to simulate real-world interview environments using text, voice, and webcam inputs. This agent dynamically captures user responses across these modalities and applies specialized NLP and computer vision techniques for feedback. For voice inputs, the system uses speech recognition to convert speech to text, which is then passed through a sentiment analysis pipeline. Webcam input is processed using the DeepFace library to detect emotions such as happiness, surprise, or neutrality in real time. The text response is also analyzed using sentiment classification models. The collected responses are then mapped to a curated CSV dataset of interview questions and ideal answers. Keyword extraction and exact matching techniques are employed to select the most relevant interview question for the user, based on their job description and career path. For evaluating the quality of responses, the system computes cosine similarity between the user's answers and predefined answers from the dataset, enabling a measurable scoring mechanism that supports personalized interview feedback. To quantitatively assess the relevance of user responses to dynamically selected interview questions, the system employs cosine similarity as the primary metric. Given a user response vector u and a reference answer vector d, from the dataset,

Question ID	Input Modality	Cosine Similarity	Interpretation
Q01	Text	0.89	Strong Semantic Match
Q01	Voice	0.83	Strong Semantic Match
Q01	Webcam	—	Emotion: Neutral
Q07	Text	0.74	Relevant Response
Q07	Voice	0.69	Moderate Match
Q07	Webcam	—	Emotion: Confident

TABLE II
MULTIMODAL INTERVIEW EVALUATION WITH COSINE SIMILARITY

this formulation captures the angular closeness between semantic vectors, offering robustness to variations in syntax or length. A threshold of 0.75 was chosen empirically to classify answers as "strong matches," supported by pilot testing. Table 2 summarizes system performance across modalities for two representative interview questions.

IV. RESULTS

The modular, agent-oriented design of the AI-driven career mentorship system yielded measurable improvements in accuracy, personalization, and system responsiveness compared to traditional monolithic or embedding-based approaches. The adoption of discrete agents—Resume, Mentor, Search, Application, and Chat—enabled a logical separation of tasks, allowing each component to be evaluated independently and optimized based on its role within the pipeline.

A. System Accuracy and Performance

Evaluation of the Resume Agent was conducted across 10 diverse resumes spanning roles in software engineering, machine learning, DevOps, and cybersecurity. Using a combination of PyMuPDF and spaCy for parsing, the system achieved over 92% accuracy in skill extraction, outperforming GPT+NER-based parsing models such as those used in Steve [2], while reducing latency and maintaining transparency. The Mentor Agent, responsible for skill-to-role mapping and fit scoring, demonstrated 93–96% accuracy when matching users to appropriate levels in a JSON-based career tree. This was validated using known ground-truth labels on 20 benchmarked resumes and compared against a cosine similarity baseline model, where the overlap-based approach consistently produced more interpretable results. The Search Agent delivered real-time job market summaries with a median latency of 1.2 seconds and a keyword mismatch error rate below 5%, demonstrating its efficiency in leveraging APIs such as SerpAPI and JSearch to contextualize career plans with live opportunities. The Application Agent generated professional-grade application bundles—including tailored resumes and personalized cover letters—rendered into PDF using PyLaTeX. This output format improved usability over JSON or plain-text exports and met formatting standards common in industry submission workflows. The Chat Interface achieved high user satisfaction during a pilot evaluation with 10 users, receiving a mean rating of 4.7/5 based on response clarity, personalization, and professional tone. Quantitatively, the chatbot maintained 98% response coherence, 95% personalization recall, and an average first token latency of 0.6 seconds when using Gemini as the LLM backbone.

B. Interview Simulation and Feedback Accuracy

Evaluation of the Multimodal Interview Agent was conducted using 15 job descriptions and corresponding candidate responses. The system achieved over 90% accuracy in mapping job description keywords to relevant interview questions. Cosine similarity scores correlated strongly ($r = 0.87$) with human evaluations of answer relevance and quality. The voice sentiment and webcam emotion analysis provided additional layers of feedback. DeepFace emotion detection achieved 86% accuracy on facial emotion classification, and sentiment analysis on voice/text reached over 93% precision in polarity detection. This multimodal pipeline resulted in richer candidate profiling and deeper insights for mentorship feedback.

V. CONCLUSION

The deployment of multi-agent systems orchestrated through LangGraph, supported by Gemini and lightweight NLP pipelines, resulted in a highly interactive, accurate, and extensible AI-driven career mentorship platform. By decoupling tasks into specialized agents, including a newly introduced Multimodal Interview Agent, the system achieves robust task-specific performance, fault tolerance, and scalability. The integration of sentiment and emotion analysis from voice and webcam input, along with cosine-based response scoring, improves the platform's ability to provide contextually rich and personalized feedback, bringing mentorship closer to real-world interview preparation scenarios. Unlike previous platforms that offered static guidance or generic templates, this system delivers contextual, skill-based, and role-specific mentorship, complete with live job insights and customized application generation. This framework sets a precedent for future AI mentorship systems to integrate LLM reasoning, deterministic logic, and real-time data aggregation in a seamless user experience. It bridges the gap between personalized career development and the dynamic labor market, offering actionable guidance to early-career professionals and students.

VI. FUTURE WORK

Future work will explore the addition of adaptive interview simulations with real-time feedback, as well as longitudinal performance tracking to refine career recommendations based on historical response patterns. The assessment component may also be expanded with interactive quizzes and mock

evaluations, updating the user's profile with performance-based insights. To further enhance role-specific mentorship, the system could incorporate hierarchical skill taxonomies and domain ontologies, enabling more granular gap analysis and better targeting for specialized roles like Prompt Engineer, MLOps Specialist, or AI Auditor. Additionally, integrating a Retrieval-Augmented Generation (RAG) module powered by a curated database of past interviews, technical assessments, and domain-specific guidance would provide richer, more context-aware support. Lastly, the currently static career tree could evolve into a dynamic, customizable structure, allowing users to generate tailored career paths based on real-time job market trends or organizational role ladders. Such upgrades would transform the system into a fully adaptive, cross-domain mentorship platform capable of supporting diverse career journeys in a rapidly evolving workforce landscape.

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